

Exercise No 2.1 -

Set builder Notation $\rightarrow$ (Form of English)

$$A = \{x | x \in \mathbb{N} \wedge x < 10\}$$

Question No 1 :-

(a) $\{x | x \text{ is a real number such that } x^2 = 1\}$

$$= \{x | x^2 = 1\}$$

$$= \{-1, 1\}$$

(b) $\{x | x \text{ is a positive integer less than } 12\}$

$$= \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$$

(c) $\{x | x \text{ is the square of an integer and } x < 100\}$

$$= 1^2 = 1$$

$$= 2^2 = 4$$

$$= 3^2 = 9$$

$$= 4^2 = 16$$

$$= 5^2 = 25$$

$$= 6^2 = 36$$

$$= 7^2 = 49$$

$$= 8^2 = 64$$

$$= 9^2 = 81$$

(d) $\{x | x \text{ is an integer such that } x^2 = 2\}$

$$= \{x | x^2 = 2\}$$

$$= \emptyset$$

~~code B~~

Question No 2 :-

Use set builder notation to give a description of each of these sets.

~~code B~~

$$\{0, 3, 6, 9, 12\}$$

Sol :-

$$\{x | x = 3n \text{ for some integer } n, 0 \leq x \leq 12\}$$

~~code B~~

$$\{-3, -2, -1, 0, 1, 2, 3\}$$

Sol :-

$$\{x | x \in \mathbb{Z}, -3 \leq x \leq 3\}$$

~~code B~~

$$\{m, n, o, p\}$$

Sol :-

$$\{x | x \in \{m, n, o, p\}\}$$

Subset :-

$$A = \{1, 2, 3, 4, 5\}$$

$$B = \{2, 5\} \quad \text{All elements, Some elements}$$

$$B \subset A$$

Question No 3 :-

For each of these pairs of sets determine whether the first is a subset of the second, the second is the subset of the first, or neither is a subset of other.

(a)

the set of airline flights from New York to New Dehli, the set of nonstop airline flights from New York to New Dehli

Set $\rightarrow$

$A = \{ \text{the set of airline flights from New York to New Dehli} \}$

$B = \{ \text{the set of nonstop airline flights from New York to New Dehli} \}$

$$B = A$$

Set of nonstop flights $\subseteq$ Set of all flights.

(b)

the set of people who speak English, the set of people who speak chinese.

$A = \{ \text{the set of people who speak English} \}$

$B = \{ \text{the set of people who speak Chinese} \}$

Neither set is a subset of the other.

(c)

the set of flying Squirrels, the set of living creatures that can fly.

$A = \{ \text{the set of flying Squirrels} \}$

$B = \{ \text{the set of living creatures that can fly} \}$

$$A = B$$

Set of flying squirrels $\subseteq$ Set of living creatures that can fly.

Question No 4 -

Q(a)
the set of people who speak English, the set of people who speak English with an Australian accent.

$A = \{ \text{the set of people who speak English} \}$
 $B = \{ \text{the set of people who speak English with an Australian accent} \}$

$$B = A$$

Set of people who speak English with Australian accent $\subseteq$ Set of people who speak English.

Q(b)
the set of fruits, the set of Citrus fruits.

$A = \{ \text{the set of fruits} \}$
 $B = \{ \text{the set of citrus fruits} \}$

$$B = A$$

Set of citrus fruits $\subset$ Set of fruits.

Q(c)
the set of students studying discrete mathematics, the set of students studying data structure.

$A = \{ \text{set of students studying discrete mathematics} \}$
 $B = \{ \text{set of students studying data structure} \}$

Different.

Neither set is a subset of others.

Set: Repeated elements are ignored.

Question No 5:-

Determine whether each of these pair of sets are equal.

Q(a)

$$\{1, 3, 3, 3, 5, 5, 5, 5, 5\}$$

First set

$$= \{1, 3, 5\}$$

Second set

$$= \{1, 3, 5\}$$

$$\{1, 3, 3, 3, 5, 5, 5, 5, 5\} = \{5, 3, 1\}$$

Q(b)

$$\{\{1\}\}, \{1, \{1\}\}$$

Because outside set 1 is not equal.

$$\{\{1\}\} \neq \{1, \{1\}\}$$

Q(c)

$$\emptyset, \{\emptyset\}$$

One set is empty and other contain empty set of an element so,

$$\emptyset \neq \{\emptyset\}$$

Question 10 6.

Suppose that $A = \{2, 4, 6\}$, $B = \{2, 6\}$,
 $C = \{4, 6\}$ and $D = \{4, 6, 8\}$ $\subseteq$

Determine which of these sets are subsets
of which other of these sets.

Given :-

$$A = \{2, 4, 6\}$$

$$B = \{2, 6\}$$

$$C = \{4, 6\}$$

$$D = \{4, 6, 8\}$$

B is a subset of A because all elements
of B are in A.

$$B \subseteq A$$

C is a subset of A because all elements
of C are in A.

$$C \subseteq A$$

A is not a subset of D because all
elements of D are not in A.

∴ False

B is not a subset of D because all elements
of D are not in B.

∴ False

C is a subset of D.

$$C \subseteq D$$

Similarly

$$B \subseteq A, C \subseteq A, C \subseteq D.$$

Question No 7:-

For each of the following sets, determine whether 2 is an element of that set.

(a)

$\{x \in \mathbb{R} \mid x \text{ is an integer greater than } 1\}$

2 is an integer greater than 1

$2 \in \{x \in \mathbb{R} \mid x \text{ is an integer greater than } 1\}$

(b)

$\{x \in \mathbb{R} \mid x \text{ is the square of an integer}\}$

2 is not a perfect square of all numbers.

$2 \notin \{x \in \mathbb{R} \mid x \text{ is the square of an integer}\}$

(c)

$\{2, \{2\}\}$

2 is explicitly listed!

$2 \in \{2, \{2\}\}$

(d)

$\{\{2\}, \{\{2\}\}\}$

$$2 \notin \{\{2\}, \{\{2\}\}\}$$

no element

$$\{\{2\}, \{2\{2\}\}\}$$

inside the sets so 2 is not an element of this set

$$2 \notin \{\{2\}, \{2\{2\}\}\}$$

no element

$$\{\{\{2\}\}\}$$

It is inside a nested set so 2 is not equal to element of this set

$$2 \notin \{\{\{2\}\}\}$$

Question 8.

For each of the sets in Exercise 7, Determine whether $\{2\}$ is an element of that set.

(a)

$$\{x \in \mathbb{R} \mid x \text{ is an integer greater than } 1\}$$

Set $\{2\}$ is not one of its element. $\{2\}$ is not the element of this set

$$2 \notin \{x \in \mathbb{R} \mid x \text{ is an integer greater than } 1\}$$

(b)

$\{x \in \mathbb{R} \mid x \text{ is the square of an element}\}$

$\{2\} \notin \{x \in \mathbb{R} \mid x \text{ is square of integer}\}$

(c)

$\{2, \{2\}\}$

$\{2\}$ is explicitly an element.

$\{2\} \in \{2, \{2\}\}$

(d)

$\{\{2\}, \{\{2\}\}\}$

$\{2\} \in \{\{2\}, \{\{2\}\}\}$

(e)

$\{\{2\}, \{2, \{2\}\}\}$

The set contains $\{2\}$ of an element.

$\{2\} \in \{\{2\}, \{2, \{2\}\}\}$

(f)

$\{\{\{\{2\}\}\}\}$

$\{2\} \notin \{\{\{\{2\}\}\}\}$

① Empty Set is the subset of all sets, $\phi \subseteq \{0\}$ ✓

② $\{\phi\} \subseteq \{\phi\}$ ✓

$\subseteq$ = Subset (some, all) $\in$ = Member

$\subset$ = Proper subset (Some elements)

Question No 9-

Determine whether each of these statement is true or false

(a)

$$0 \in \phi$$

False

(b)

$$\phi \in \{0\}$$

False

(c)

$$\{0\} \subset \phi$$

False

$$A = \{0, 1, 2, 3\}$$

$$\{0\} \in B = \{\phi, \{\phi\}\}$$

(d)

$$\phi \subset \{0\}$$

Empty set is a subset of every set

True

(e)

$$\{0\} \in \{0\}$$

~~True~~ False

$$(f) \quad \{0\} \subset \{0\}$$

False

(g)

$$\{\emptyset\} \subseteq \{\emptyset\}$$

Complete subset means sum or all elements
True.

Question No 10:-

Determine whether these statements are true or false

(a)

$$\emptyset \in \{\emptyset\}$$

True.

(b)

$$\emptyset \in \{\emptyset, \{\emptyset\}\}$$

True.

(c)

$$\{\emptyset\} \in \{\emptyset\}$$

False.

(d)

$$\{\emptyset\} \in \{\{\emptyset\}\}$$

True

(e)

$$\{\emptyset\} \subset \{\emptyset, \{\emptyset\}\}$$

True.

(f)

$$\{\{\emptyset\}\} \subset \{\emptyset, \{\emptyset\}\}$$

Some elements

True.

(g)

$$\{\{\emptyset\}\} \subset \{\{\emptyset\}, \{\emptyset\}\}$$

True.

Question 2011 :-

Determine whether each of these statements is true or false.

(a)

$$x \in \{x\}$$

True

(b)

$$\{x\} \subseteq \{x\}$$

True

(c)

$$\{x\} \in \{x\} \rightarrow \{x, \{x\}\}$$

False.

(d)

$$\{x\} \in \{\{x\}\}$$

True

(e)

$$\emptyset \subseteq \{x\}$$

Empty set har kisi ka subset hota ha.

True

↳ Proper ma b

↳ Subset: ma b.

(f)

$$\emptyset \in \{x\}$$

False.

Venn Diagram:-

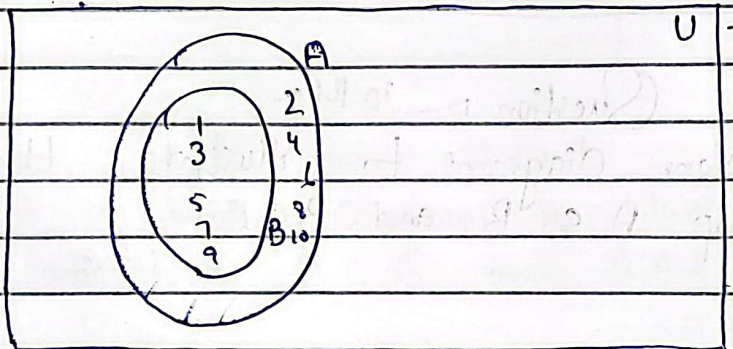
Universal set :- All elements

Question No 12:-

Use a Venn diagram to illustrate the subset of odd integers in the set of all positive integers not exceeding 10.

Sol:- (0-10) integers
Pick up odd integers

1, 3, 5, 7, 9



$$A = \{1, 2, 3, 4, \dots, 10\}$$

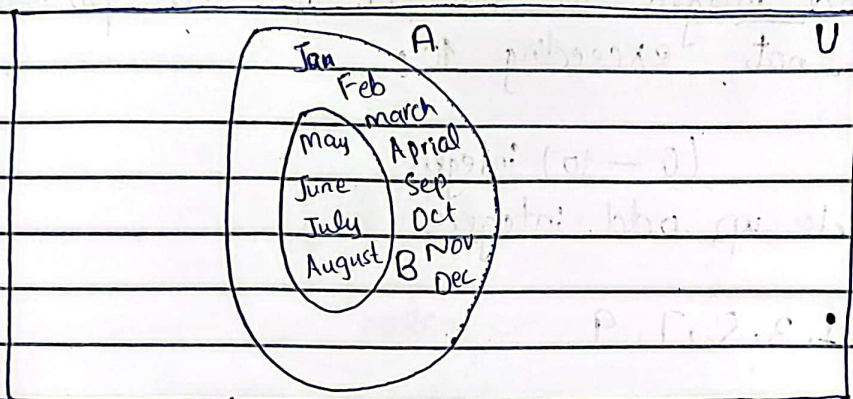
$$B = \{1, 3, 5, 7, 9\}$$

Question No 13:-

Use a Venn diagram to illustrate the set of all months of the year whose names do not contain the letter R in the set of all months of the year.

$A = \{ \text{January, February, March, April, May, June, July, August, September, October, November, December} \}$

$B = \{ \text{May, June, July, August} \}$



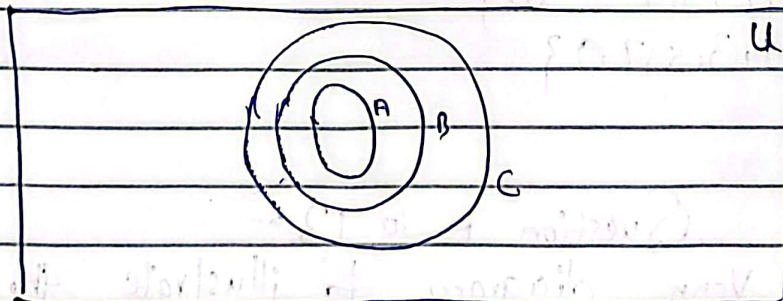
Question 10/14:-

Use a Venn diagram to illustrate the relationship $A \subseteq B$ and $B \subseteq C$.

$A = \{ 1, 2, 3, 4 \}$

$B = \{ 1, 2, 3, 4, 5 \}$

$C = \{ 1, 2, 3, 4, 5, 6, 7 \}$



Question No 15 :-

Use a Venn diagram to illustrate the relationships $A \subset B$ and $B \subset C$.

Sol :-

$$A = \{3, 4, 5, 6\}$$

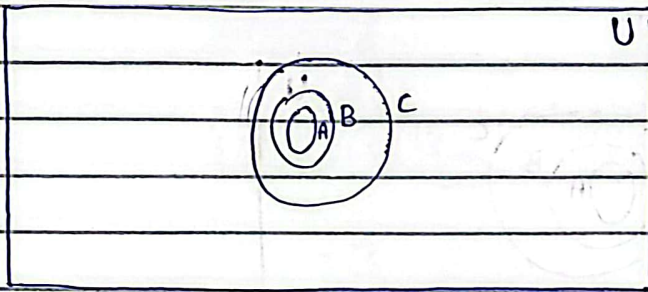
$$B = \{5, 6\}$$

$$C = \{6\}$$

$$B = \{3, 4, 5, 6\}$$

$$A = \{5, 6\}$$

$$C = \{6, 3, 4, 5, 8\}$$



Question No 16 :-

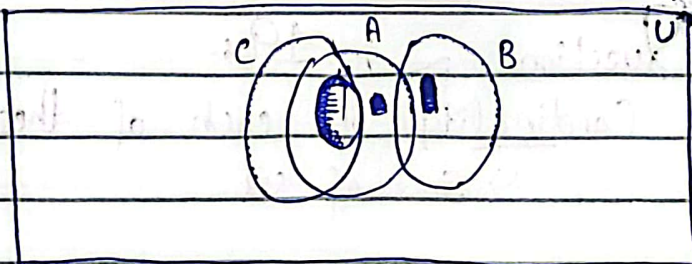
Use a Venn diagram to illustrate the relationships $A \subset B$ and $A \subset C$.

Sol :-

$$B = \{4, 5, 6, 7, 8, 9\}$$

$$A = \{7, 8\}$$

$$C = \{7, 8, 9, 10\}$$



Question No 17:-

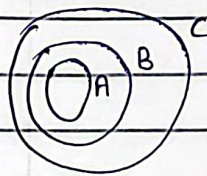
Suppose that A, B and C are sets such that $A \subseteq B$ and $B \subseteq C$. Show that $A \subseteq C$.

Sol:-

$$B = \{6, 7, 8, 9\}$$

$$A = \{9, 7\}$$

$$C = \{6, 7, 8, 9, 2, 3\}$$



Question No 18:-

Find two sets A and B such that $A \in B$ and $A \subseteq B$.

Sol:-

$$A = \{x\}$$

$$B = \{\{x\}, x\}$$

Question No 19:-

What is the Cardinality of each of these sets?
↳ Size of set

(a)

$$\{a\}$$

Sol:-

$$\{a\} = 1$$

(b)

$$\{\{a\}\}$$

$$= 1$$

(c)

$$\{a, \{a\}\}$$

$$= 2$$

(d)

$$\{a, \{a\}, \{\{a, \{a\}\}\}$$

$$= 3$$

Question No 20:-

What is the Cardinality of each of these sets?

$$(a)$$

$$\emptyset$$

$$= 0$$

(b)

$$\{\emptyset\}$$

$$= 1$$

(c)

$$\{\emptyset, \{\emptyset\}\}$$

$$= 2$$

$$\{ \emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\} \}$$

$$= 3$$

Power set :- Formula :- 2^n $\emptyset$ Compulsory

Question No 21 :-

Find the power set of each of these sets.
where a and b are distinct elements.

(a)

$$\{\emptyset, \{a\}\}$$

$$2^n = 2^1 = 2$$

$$P = \{\emptyset, \{a\}\}$$

(b)

$$\{a, b\}$$

$$2^n = 2^2 = 4$$

$$P = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

(c)

$$\{\emptyset, \{\emptyset\}\}$$

$$= 2^n = 2^2 = 4$$

$$P' = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$$

Elements :-

$$2^n$$

Question No 23 :-

How many elements does each of these sets have where A and B are distinct elements?

(a)

$$P(a, b, \{a, b\})$$

$$2^n = 2^3 = 8$$

Soln -

$$(b) \quad P(\{\phi, a, \{a\}, \{\{a\}\}\})$$

$$2^n = 2^4 = \boxed{16}.$$

$$(c) \quad P(P(\phi))$$

$$2^n = 2^2 = \boxed{4}.$$

Question 10 Q48 -
Determine whether each of these sets is the power set of a set, where a and b are distinct elements.

(a)

ϕ

NO its not a powerset of any set.

(b)

$$\{\phi, \{a\}\}$$

Yes, it is power set of set $= \{a\}$

(c)

$$\{\phi, \{a\}, \{\phi, a\}\}$$

NO it is not a power set of set.

(d)

$$\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

Yes, ~~$\{a\} = \{b\}$~~
it is a power set of set $\{a, b\}$

Question ID 28:-

What is the Cartesian product $A \times B$, where A is the set of courses offered by the mathematics department at a University and B is the set of mathematics professors at this University? Give an example of how this Cartesian product can be used.

$A = \{ \text{Set of Courses} \}$ offered by m

~~$B = \{ \text{Set of mathematical } x \}$~~
 ~~$B = \{ \text{Set of mathematics professor} \}$~~
 $B = \{ \text{Set of mathematics professor} \}$

$A = \{ \text{Discrete, Calculus, Statics} \}$

$B = \{ \text{Dr. Nadeem, Dr. Faisal, Sir Hamza} \}$

$A \times B = \{ \{ \text{Discrete, Dr. Nadeem} \}, \{ \text{Discrete, Dr. Faisal} \}, \{ \text{Discrete, Sir Hamza} \}, \{ \text{Calculus, Dr. Nadeem} \}, \{ \text{Calculus, Dr. Faisal} \}, \{ \text{Calculus, Sir Hamza} \}, \{ \text{Statics, Dr. Nadeem} \}, \{ \text{Statics, Dr. Faisal} \}, \{ \text{Statics, Sir Hamza} \} \}$